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ASSIGNMENT 2	
GRADUATE SCHOOL	
PROGRAMME: Master of Science in Information Technology	
MODULE CODE AND TITLE: MIT91207: Machine Learning	LEVEL AND TRIMESTER: LEVEL 9, TRIMESTER 4
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LECTURER: DR. GUSTAVE UDAHEMUKA	TOTAL MARKS: 80 TO BE SCALED TO 15 MARKS

QUESTION 1 (10 Marks)

A machine learning engineer develops a Decision Tree Classifier to predict whether customers will leave a telecommunications service.

The engineer uses the following configuration:

- `max_depth = None`
- `min_samples_leaf = 1`

The resulting model achieves 99.6% accuracy on the training set, but only 59.4% accuracy on the validation set.

Required:

(a) Analyze the difference between the training and validation accuracies. Identify the most likely cause of this performance gap and explain the evidence supporting your conclusion. **(3 Marks)**

(b) Explain the roles of bias and variance in this situation. Based on the reported results, determine whether the model exhibits characteristics of high bias or high variance, and justify your answer. **(3 Marks)**

(c) Explain how the specified values of *max_depth* and *min_samples_leaf* affect the complexity of the Decision Tree. **(2 Marks)**

(d) Explain the difference between overfitting and underfitting. For the model described above, explain which condition is present and describe two changes to the Decision Tree configuration that could help improve its generalization performance. **(2 Marks)**

QUESTION 2 (10 Marks)

A financial technology company evaluates two classification models for detecting fraudulent transactions: a parametric Linear SVM and a non-parametric Decision Tree Classifier. Both models are evaluated using the same training and validation data.

The following results are obtained:

Model	Training F1-score	Validation F1-score
Linear SVM	0.79	0.76
Decision Tree	0.99	0.63

Required:

(a) Analyze the validation performance of the two models. Explain how a parametric model and a non-parametric model differ in the way they represent relationships in data and discuss how these structural differences influence their inductive biases.

Based on the reported training and validation F1-scores, explain which model exhibits stronger evidence of overfitting and why. **(5 Marks)**

(b) Identify three specific Decision Tree hyperparameters that could be included in a hyperparameter optimization grid to reduce overfitting. **(5 Marks)**

For each hyperparameter:

1. explain what the hyperparameter controls;
2. describe how changing its value affects the structure or growth of the tree; and
3. explain how this change can help control model complexity and improve generalization.

QUESTION 3 (10 Marks)

A machine learning practitioner is developing a binary classification model. To improve the model, the practitioner evaluates 50 different hyperparameter combinations using the same fixed validation dataset.

After testing all 50 combinations, the practitioner selects the configuration that produces the highest validation score and reports this score as an estimate of the model's expected performance on new, unseen data.

Required:

(a) Explain how cross-validation should be incorporated into the hyperparameter tuning process.

Why can repeatedly evaluating many different hyperparameter combinations on the same fixed validation dataset lead to an over-optimistic estimate of generalization performance?

Describe an appropriate procedure for separating hyperparameter selection from final model evaluation. **(5 Marks)**

(b) Explain how K-Fold Cross-Validation works when evaluating and selecting a machine learning model.

Then compare K-Fold Cross-Validation with Stratified K-Fold Cross-Validation for a binary classification problem in which the minority class represents only 11% of the observations.

Explain why maintaining approximately the same proportion of each class across the folds is important in this situation. **(5 Marks)**

QUESTION 4 (10 Marks)

A data operations team is developing a fraud-detection system and wants to automate model tuning while avoiding data leakage and lookahead bias.

Required:

(a) Assume that a dataset contains 2% fraudulent transactions and 98% legitimate transactions.

Demonstrate mathematically how a classifier that predicts every transaction as Legitimate can achieve a high classification accuracy while failing completely to detect fraud.

Explain why accuracy alone can therefore be misleading when evaluating highly imbalanced classification problems, particularly for an early-warning fraud detection system. **(3 marks)**

(b) Using the Credit Card Fraud Detection Dataset provided in the file *credit.csv*, write a clean and reproducible Python script that implements an appropriate machine learning workflow. **(7 marks)**

Your script must sequential:

1. separate the predictor variables from the binary fraud target variable;
2. divide the data into training and testing sets using *train_test_split* with stratification;
3. configure a baseline Decision Tree Classifier with an explicit *random_state*;
4. define a hyperparameter grid containing at least the two complexity-related parameters *max_depth* and *min_samples_leaf*;
5. perform cross-validated hyperparameter optimization using *GridSearchCV* on the training data only; and
6. evaluate the final selected model on the held-out test set using exactly two classification metrics appropriate for an imbalanced fraud-detection problem.

QUESTION 5 (10 Marks)

A credit-risk team uses the Home Credit Default Risk Dataset (provided in the file *home.csv*) to predict whether loan applicants are likely to experience repayment difficulties.

The dataset contains numerical variables such as *AMT_INCOME_TOTAL* and *DAYS_EMPLOYED*, a high-cardinality categorical variable such as *ORGANIZATION_TYPE*, and an ordinal variable such as *NAME_EDUCATION_TYPE*. The data also contain missing values, while numerical variables have substantially different ranges.

Required:

(a) Design a comprehensive preprocessing strategy for this heterogeneous dataset.

Specify appropriate methods for:

- handling missing numerical and categorical values;
- encoding high-cardinality nominal variables;
- encoding ordinal variables; and
- scaling numerical features.

Explain how inappropriate encoding or scaling decisions can affect the performance of tree-based and distance-based classification algorithms. **(3 Marks)**

(b) Explain the mathematical mechanism by which calculating imputation statistics, such as medians, or scaling parameters using the entire dataset before cross-validation can introduce data leakage.

Then explain how a Scikit-Learn *Pipeline* combined with *ColumnTransformer* prevents this form of leakage during model training and cross-validation. **(3 Marks)**

(c) Write a complete and reproducible Python script that: **(4 marks)**

1. automatically identifies numerical and categorical features;

2. creates separate preprocessing procedures for the two feature types;
3. performs appropriate missing-value imputation;
4. encodes categorical variables;
5. scales numerical variables where appropriate;
6. combines the preprocessing operations using *ColumnTransformer*;
7. places a classifier inside a parent Pipeline; and
8. evaluates the complete pipeline on held-out test data.

QUESTION 6 (10 Marks)

An online transaction monitoring system has a background fraud rate of approximately 3%. At its current classification threshold of $\tau = 0.50$, the system produces the following confusion matrix:

	Predicted Fraudulent	Predicted Legitimate
Actual Fraudulent	120	30
Actual Legitimate	450	14,400

Required:

(a) Using the confusion matrix, calculate the following metrics: **(3 marks)**

- precision;
- recall (sensitivity);
- specificity (which is given as $TN/(TN + FP)$); and
- F_1 -score.

Show the mathematical calculation for each metric and briefly interpret what each metric indicates in the context of fraud detection.

(b) Suppose the classification threshold is increased from $\tau = 0.50$ to $\tau = 0.80$. **(2 marks)**

Explain the expected effect of this change on:

- precision;
- recall;
- false positives; and
- false negatives.

Explain how the relative costs of false positives and false negatives should be considered when selecting an operational threshold.

(c) Compare the analytical purposes of the Receiver Operating Characteristic (ROC) curve and the Precision-Recall (PR) curve.

Explain why the PR curve can provide more informative performance analysis than the ROC curve for highly imbalanced classification problems. **(2 marks)**

(d) Write a reproducible Python code snippet that: **(3 marks)**

1. obtains predicted probabilities from a trained classifier using `.predict_proba()`;
2. applies a custom probability threshold to generate binary predictions; and
3. calculates two appropriate evaluation metrics.

Then describe how a machine learning practitioner could systematically evaluate a range of threshold values using a loop over an array of thresholds to identify a suitable operational decision boundary.

QUESTION 7 (10 Marks)

A property-fintech company uses the House Prices: Advanced Regression Techniques Dataset (provided in the file *prices.csv*) to predict the continuous target variable *SalePrice*.

Exploratory analysis indicates substantial multicollinearity among some predictors. For example, *GrLivArea* is strongly related to *TotRmsAbvGrd*, while *GarageCars* is strongly related to *GarageArea*. The dataset also contains several potentially uninformative features.

The company wants to compare Ordinary Least Squares (OLS), Ridge Regression, and Lasso Regression.

Required:

(a) Design a systematic experimental procedure for comparing OLS, Ridge, and Lasso Regression. **(2 Marks)**

Your procedure should address:

- data partitioning;
- feature preprocessing and scaling;
- cross-validated hyperparameter tuning; and
- final evaluation using previously unseen test data.

(b) Explain how strong correlations among predictors can affect an unregularized OLS model.

Compare the effects of Ridge Regression (L2 regularization) and Lasso Regression (L1 regularization) on multicollinearity.

Explain the mathematical mechanism by which Lasso can perform automatic feature selection. **(3 Marks)**

(c) Write a Python script that: **(3 marks)**

1. partitions the dataset into training and test sets;
2. applies appropriate feature scaling;
3. fits at least two of the three regression models;
4. generates predictions on the test set; and
5. reports the exact values of MAE, RMSE, and R^2 for each fitted model.

(d) Suppose the following test-set results are obtained: **(2 marks)**

Model	R^2
OLS	0.70
Ridge	0.76
Lasso	0.74

Interpret these results and explain what additional considerations, such as model interpretability, feature selection, regularization, and model complexity, should be examined before selecting a model for deployment.

QUESTION 8 (10 Marks)

A company wants to divide a large set of customers into groups based on their purchasing and usage behaviour. Since the dataset does not contain predefined class labels, the company decides to use K-Means Clustering.

As an initial experiment, K-Means is applied directly to the raw, unstandardized features. One feature has values ranging from 0 to approximately 20,000, while another feature ranges from 0 to 1. The resulting model assigns more than 95% of the customers to a single cluster.

Required:

(a) Identify and explain four technical factors that could contribute to the highly unbalanced clustering result. **(3 Marks)**

Your answer should consider:

1. differences in feature scales;
2. inappropriate or weakly informative features;
3. the influence of outliers; and
4. an inappropriate choice of the number of clusters, k .

(b) Design a systematic workflow for applying K-Means Clustering to this customer dataset. **(3 marks)**

Your workflow should describe the main steps from raw data to final cluster interpretation, including:

- data inspection;
- missing-value handling;
- feature selection;
- treatment of outliers;
- feature standardization;
- selection of the number of clusters;
- K-Means model fitting; and
- interpretation and profiling of the resulting clusters.

(c) Explain how the Elbow Method using inertia and the Silhouette Score can be used to evaluate different values of k .

Suppose the analysis produces a maximum Silhouette Score of 0.56 at $k = 4$, while the inertia curve shows a clear point of diminishing improvement at $k = 3$.

Explain how these two results should be considered when determining an appropriate number of clusters. **(2 marks)**

(d) Explain why a mathematically distinct cluster is not necessarily a meaningful business segment.

Describe two types of additional evidence or domain knowledge that should be considered before using the resulting clusters to design customer groups or business strategies. **(2 marks)**

END
